

# Conjecture 00000005600: same determinant, non-isomorphic echelon forms, separated by a noncommutative unimodular pair

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## Statement

*English.* Definition: The determinant and the echelon form are two layers. Conjecture: There exist two matrices with the same determinant but non-isomorphic echelon forms, and the separation is realized by an explicit noncommutative pair of unimodular transformations. (The Chinese text says the same: the two matrices have equal determinants while their echelon forms are not isomorphic, and the separation is realized by an explicit noncommutative pair of unimodular transformations.)

**Verdict: proved.** The statement is existential; we give explicit integer matrices and an explicit pair in  $\text{GL}_2(\mathbb{Z})$ , and the whole statement below is a single Lean theorem, `Conjecture5600.conjecture_5600`, checked against Mathlib.

## 1 Definitions and reading

All matrices are  $2 \times 2$  integer matrices. *Unimodular transformations* are the elements of  $\text{GL}_2(\mathbb{Z})$  (integer matrices with integer inverse, equivalently  $\det = \pm 1$ ); in Lean this is Mathlib's `GL (Fin 2) ℤ`. A unimodular row transformation sends  $A$  to  $WA$ ,  $W \in \text{GL}_2(\mathbb{Z})$ ;  $A$  and  $H$  are *row equivalent* if  $H = WA$  for some such  $W$ .

*Echelon form.* Over  $\mathbb{Z}$  the echelon form of a nonsingular matrix is its row Hermite normal form:  $H = WA$  with  $W \in \text{GL}_2(\mathbb{Z})$ ,  $H$  upper triangular, positive diagonal (pivot) entries, and every entry above a pivot in  $[0, \text{pivot})$ . This is the Lean predicate `IsHermiteNormalForm` (stated for all  $n \times n$ ).

*Isomorphism type of an echelon form.* The row lattice  $L(A) = \{wA : w \in \mathbb{Z}^2\}$  (Mathlib: `LinearMap.range A.vecMulLinear`, proved equal to the  $\mathbb{Z}$ -span of the rows) is unchanged by row transformations, so all echelon forms of  $A$  have the same row lattice and the same cokernel  $\text{coker } A = \mathbb{Z}^2 / L(A)$ . We say two echelon forms are *non-isomorphic* in the strongest sense: their cokernels are not isomorphic as abelian groups. This implies that they are different matrices, that the matrices are not row equivalent, and that their Smith forms differ.

*The separating pair.* We prove two readings, with the same explicit pair

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad UV = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq VU = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

the two standard elementary generators of  $\text{SL}_2(\mathbb{Z})$ , with  $\det U = \det V = 1$ .

- (A) *The pair carries the two matrices to their separated echelon forms:*  $UM$  and  $VN$  are the Hermite forms of  $M$  and  $N$ ,  $\det M = \det N$ , and the echelon forms are non-isomorphic.

- (B) *The separation comes from the noncommutativity itself:* for one base matrix  $X$ , the matrices  $XUV$  and  $XVU$  (the pair applied in the two orders) have equal determinants, since  $\det(UV) = \det(VU)$ , but distinct echelon forms.

## 2 Theorem and proof

Let

$$M = \begin{pmatrix} 1 & -4 \\ 0 & 4 \end{pmatrix}, \quad N = \begin{pmatrix} 2 & 0 \\ -2 & 2 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

**Theorem 1.** 1.  $\det M = \det N = 4$ ,  $UM = \text{diag}(1, 4)$  and  $VN = \text{diag}(2, 2)$  are in Hermite normal form, and  $UM \neq VN$ .

2. For all matrices  $H_1$  row equivalent to  $M$  and  $H_2$  row equivalent to  $N$  (in particular for all echelon forms of  $M$  and  $N$ , under any convention),  $\text{coker } H_1 \not\cong \text{coker } H_2$  and  $H_1 \neq H_2$ .
3. There are no  $P, Q \in \text{GL}_2(\mathbb{Z})$  with  $PNQ = M$ ; so  $M$  and  $N$  are not even equivalent under two-sided unimodular transformations, and their Smith forms differ.
4.  $\det(XUV) = \det(XVU) = 2$ ;  $XUV$  has Hermite form  $\text{diag}(2, 1)$  and  $XVU$  has Hermite form  $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ ; and every matrix row equivalent to  $XUV$  differs from every matrix row equivalent to  $XVU$ .

*Proof.* (1) Direct multiplication:  $\det M = 4 - 0 = 4$ ,  $\det N = 4 - 0 = 4$ ,  $UM = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$ ,  $VN = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ . Both are upper triangular with positive diagonal and zero above-pivot entries, so they are Hermite forms; their  $(1, 1)$  entries  $1 \neq 2$  differ.

(2) If  $H = WA$  with  $W \in \text{GL}_2(\mathbb{Z})$ , then  $wH = (wW)A$  and  $wA = (wW^{-1})H$ , so  $L(H) = L(A)$ . Now  $L(M) = \{(a, -4a + 4b)\} = \mathbb{Z} \times 4\mathbb{Z}$  and  $L(N) = \{(2a - 2b, 2b)\} = 2\mathbb{Z} \times 2\mathbb{Z}$ . Every class  $y$  in  $\mathbb{Z}^2/L(N)$  satisfies  $y + y = 0$ , because  $2v \in 2\mathbb{Z} \times 2\mathbb{Z}$  for every  $v$ . In  $\mathbb{Z}^2/L(M)$  the class  $x$  of  $(0, 1)$  satisfies  $x + x \neq 0$ , because  $(0, 2) \notin \mathbb{Z} \times 4\mathbb{Z}$ . An additive isomorphism  $e$  would give  $e(x + x) = e(x) + e(x) = 0$ , hence  $x + x = 0$ , a contradiction. (The cokernels are  $\mathbb{Z}/4$  and  $(\mathbb{Z}/2)^2$ .) Equal matrices have equal cokernels, so also  $H_1 \neq H_2$ .

(3) All entries of  $N$  are even, so all entries of  $PNQ$  are even for all integer  $P, Q$ ; explicitly  $(PNQ)_{11} = 2((P_{11} - P_{12})Q_{11} + P_{12}Q_{21})$ , while  $M_{11} = 1$ .

(4)  $XUV = \begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}$  and  $XVU = \begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix}$ , both of determinant 2. With  $W_1 = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$  and  $W_2 = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$  (determinant 1) we get  $W_1XUV = \text{diag}(2, 1)$  and  $W_2XVU = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ , both Hermite forms. Every vector of  $L(XUV) = \{(2a + 2b, a + 2b)\}$  has even first coordinate, while  $(1, 1)$  is a row of  $XVU$ . By the invariance of  $L$  under row transformations, a matrix row equivalent to both would give  $L(XUV) = L(XVU)$ , which is false.  $\square$

*Remark 2.* In reading (B) both matrices are obtained from the same  $X$  by unimodular column operations, so they are two-sided equivalent and their cokernels are necessarily isomorphic for every choice of  $X, U, V$ . In that reading, the layer that the determinant fails to see is the row-equivalence class, which the echelon form classifies; part (4) proves that these classes differ. Part (2) gives the cokernel form of the separation for reading (A). The conjecture is existential, and the theorem satisfies both readings with the same explicit noncommutative pair.

### 3 Lean formalization

File `lean/Conjecture5600/Basic.lean` (Lean 4.33.1, Mathlib v4.33.1; the root file `Conjecture5600.lean` only imports it). Main theorem:

```
theorem conjecture_5600 :
  exists (A B : Mat) (U V : GL (Fin 2) Int),
    (U : Mat).det = 1 /\ (V : Mat).det = 1 /\ U * V != V * U /\
    A.det = B.det /\
    IsHermiteNormalForm ((U : Mat) * A) /\ IsHermiteNormalForm ((V : Mat) * B) /\
    (U : Mat) * A != (V : Mat) * B /\
    (forall H1 H2, RowEquiv A H1 -> RowEquiv B H2 -> H1 != H2) /\
    (forall H1 H2, RowEquiv A H1 -> RowEquiv B H2 ->
      IsEmpty (coker H1 ~+ coker H2)) /\
    (forall P Q : GL (Fin 2) Int, (P : Mat) * B * Q != A) /\
    exists Y : Mat, (Y * U * V).det = (Y * V * U).det /\
      (forall H1 H2, RowEquiv (Y*U*V) H1 -> RowEquiv (Y*V*U) H2 -> H1 != H2) /\
      (exists H1 H2, RowEquiv (Y*U*V) H1 /\ IsHermiteNormalForm H1 /\
        RowEquiv (Y*V*U) H2 /\ IsHermiteNormalForm H2)
```

(ASCII transcription; the source uses the Unicode symbols, `Mat` is `Matrix (Fin 2) (Fin 2) Int`, `~+` is Mathlib's additive equivalence `AddEquiv`, and  $YUV$  is the product of matrices with  $U, V$  coerced from `GL`.) Correspondence: `det` is Mathlib's `Matrix.det`; unimodular transformations are Mathlib's `GL (Fin 2) Int`, with noncommutativity stated in the group; `RowEquiv A H` is  $\exists W \in \text{GL}_2(\mathbb{Z}), WA = H$ ; `coker X` is the quotient module  $\mathbb{Z}^2/L(X)$  with `rowModule X = L(X)` and `rowModule_eq_span` identifying it with the span of the rows. The witnesses are  $A = M$ ,  $B = N$ ,  $Y = X$  and the pair  $U, V$  above.

Checks: `lake build` succeeds, `lake env lean -DwarningAsError=true` on the file succeeds, there is no `sorry` or `native_decide`, and `#print axioms Conjecture5600.conjecture_5600` reports only `propext`, `Classical.choice`, `Quot.sound`.

### Credit

The pair  $\text{diag}(1, 4)$ ,  $\text{diag}(2, 2)$  and the parity obstruction in part (3) are adapted from the accepted solution of the near-duplicate conjecture 00000006330 by gaochengzhecpu ([solutions/00000006330/gaochengzhecpu\\_submission\\_20261003215646/](https://github.com/solutions/00000006330/gaochengzhecpu_submission_20261003215646/)), used under the repository's GPL-3.0 license. The Mathlib formalization, the cokernel invariant, the unimodular pair and reading (B) are new here.